

SOC INCIDENT ANALYSIS REPORT

ANALYST
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LAB
Oski Lab —
CyberDefenders

DATE (UTC)
Sep 28, 2022 —
17:40:46

SEVERITY
HIGH

VERDICT
TRUE POSITIVE

01 ALERT DETAILS

Alert Name	Oski Stealer (Stealc Malware Family) — Malicious Email Attachment Campaign
Source	171.22.28.221 (C2 Server)
Destination	C:\Users\admin\
Severity	HIGH

02 EXECUTIVE SUMMARY

A suspicious email containing a malicious attachment was received, triggering a security alert for a potentially harmful download linked to the **Stealc / Oski malware family**. Upon execution, the malware established communication with a C2 server at **171.22.28.221**, stole credentials from browser password stores using **sqlite3.dll**, and then self-deleted after **5 seconds** to evade forensic detection. This is a confirmed **True Positive** requiring immediate containment.

03 INVESTIGATION STEPS

- 1 Identified malware family** — Analysed the suspicious email attachment and confirmed it belongs to the Stealc/Oski malware family using VirusTotal.
- 2 Checked compilation timestamp** — Used VirusTotal to find the PE header creation time: September 28, 2022 at 17:40:46 UTC.
- 3 Identified C2 server** — Examined memory patterns and network connections in Any.run sandbox; found malware communicating with IP 171.22.28.221 via URLs ending in .php and .exe.
- 4 Analysed post-infection behaviour** — Reviewed Any.run sandbox report; identified first library loaded post-infection as sqlite3.dll, indicating credential harvesting intent.
- 5 Extracted malware configuration** — Found RC4 encryption key 5329514621441247975720749009 used to decrypt base64-encoded strings within the malware.
- 6 Mapped MITRE ATT&CK; techniques** — Identified T1555 (Credentials from Password Stores) as the primary technique used for credential theft from browsers.
- 7 Analysed child processes** — Observed cmd.exe commands targeting C:\Users\admin\AppData\Local\Temp\VPN.exe and C:\ProgramData\ for DLL deletion and self-cleanup.
- 8 Confirmed self-deletion timing** — Determined the malware deletes itself 5 seconds after completing data exfiltration.

04 KEY FINDINGS & INDICATORS OF COMPROMISE

IP 171.22.28.221 (C2 Server)

DOMAIN	dns.msftncsi.com
HASH	12c1842c3ccafe7408c23ebf292ee3d9
KEY	RC4 key: 5329514621441247975720749009

Suspicious Patterns Observed

- Malware delivered via suspicious email attachment masquerading as a legitimate file
- Immediate outbound connection to C2 server 171.22.28.221 post-execution
- Loading of sqlite3.dll — targeting browser credential databases
- Use of RC4 encryption to obfuscate configuration and strings
- Use of cmd.exe to delete itself and all associated DLL files
- Self-deletion 5 seconds post-exfiltration to hinder forensic investigation

05 ANALYSIS & REASONING

The malware follows a classic **infostealer kill chain**. It arrives via a phishing email, executes silently, immediately contacts its C2 server for instructions, then loads **sqlite3.dll** to access browser databases where credentials are stored. It uses RC4 encryption to hide its configuration from static analysis tools. After exfiltrating the stolen credentials to **171.22.28.221**, it executes a timed self-deletion command via cmd.exe to remove all traces within 5 seconds — a clear attempt to frustrate incident responders. The alignment with **MITRE T1555** confirms credential theft as the primary objective.

Alternative Explanations Ruled Out

- **Legitimate software using sqlite3.dll** — Ruled out: file arrived via suspicious email and made outbound C2 connections before loading the DLL.
- **False positive network alert** — Ruled out: IP 171.22.28.221 is confirmed malicious via VirusTotal and Any.run; traffic pattern (.php/.exe paths) consistent with C2.
- **Accidental file deletion** — Ruled out: deletion triggered programmatically via cmd.exe with a deliberate 5-second delay, consistent with evasion behaviour.

06 VERDICT

TRUE POSITIVE

Confidence: **HIGH**
Clear evidence of malicious activity

07 RECOMMENDATIONS

Immediate Actions

- Block IP 171.22.28.221 at the firewall immediately
- Isolate the affected endpoint from the network
- Reset all credentials for the affected user — assume all browser-stored passwords are compromised
- Scan all systems for Stealc/Oski malware IOCs
- Recover residual forensic artifacts from C:\Users\admin\AppData\Local\Temp\ and C:\ProgramData\

Long-Term / Preventive Measures

- Enforce email filtering to block suspicious attachments (especially .ppt files with macros)
- Deploy endpoint detection rules to alert on sqlite3.dll being loaded by unexpected processes
- Educate users on phishing email awareness
- Implement a password manager instead of relying on browser-stored credentials

- Enable network monitoring to detect unexpected outbound connections to unknown IPs

08 LESSONS LEARNED

- Learned how to use VirusTotal and Any.run sandbox together to build a complete picture of malware behaviour
- Understood how Stealc/Oski infostealers operate — from delivery to exfiltration to self-deletion
- Practised mapping real malware behaviour to MITRE ATT&CK; techniques (T1555)
- Gained hands-on experience identifying RC4 encryption keys in malware configurations
- Learned the importance of analysing child processes in sandbox reports to uncover evasion techniques